Jinyu Zheng

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Education

New York University Expected *May 2028*
B.S. in Electrical Engineering

- GPA: 3.40/4.00
- **Current Coursework:** DSP I (Graduate), DSP Lab (Graduate), RF Communications Circuits & Components (Graduate), Image & Video Processing (Graduate), Machine Listening (Graduate)
- **Selected Coursework:** Data Structures & Algorithms, Electronics I, PCB Design, Embedded Systems, Signals & Systems, Machine Learning

Projects

STM32F405 Dev Board - *Altium Designer, PCB Design, Power Electronics*

- Designed an STM32F405 ARM Cortex-M4 development board in Altium Designer integrating SPI/I^2C peripherals and SWD debugging.
- Implemented a 24V to 3.3V dual-stage power delivery network (buck converter + LDO) featuring power multiplexing, TVS ESD, and reverse polarity protection.
- Configured PCB layer stackup and routed an impedance-matched USB-C interface using differential pair length matching, continuous ground planes, and via stitching.
- Executed DRC/DFM checks, generated Gerber/BOM files for JLCPCB fabrication.

Water Level Controller - *Bare-metal C++, Embedded System, Controls*

- Developed bare-metal C++ firmware for an ATmega32U4 MCU to operate a real-time, closed-loop physical control system.
- Programmed microcontroller registers directly to establish precise 1kHz Fast PWM control signals and configure continuous analog-to-digital conversions.
- Mitigated actuator chattering and hardware overshoot by engineering a hybrid Proportional/Hysteresis control loop and mathematically modeling physical transition delays.

Ragdoll Simulator - *Python, Physics, Modeling* Glowscript Link

- Engineered a Python/VPython physics engine simulating gravity, rotational kinematics, and momentum in real time for an interactive rag-doll simulator.
- Implemented custom collision detection and environmental physics, including bouncing and friction to ensure mathematically accurate object interactions.

Experience

Audio Visual Technician @ KAPCQ CM *Sept 2020 - Present*

- Configured microphones, speaker systems, and wireless transmitters for live events and presentations.
- Operated multi-channel audio mixers, balancing signal levels and preventing clipping, distortion, and feedback during live operation.
- Diagnosed signal issues like interference, noise, and signal loss in audio systems during events.
- Ensured system readiness by verifying cabling, signal routing, and equipment functionality prior to events.

Skills

Languages: Python, Java, C, C++, C#, MATLAB
Software: Altium Designer, Git, Cadence Virtuoso
Lab Hardware: Oscilloscope, DMM, Bench Power Supply